STAT2056 Midterm 2 – Spring 2023

Question 1 – 30 pts

Suppose that the error in the reaction temperature, in Celsius, for a controlled laboratory experiment is a continuous random variable X having the probability density function

$$f(x) = \begin{cases} cx^2 & 1 < x < 2 \\ 0 & elsewhere \end{cases}$$

(a) (5 pts) Find the value of the constant c.
(b) (5 pts) Compute the cumulative distribution function (CDF) of X.
(c) (5 pts) Find $P(1.25 < X < 1.5)$.
(d) (5 pts) Compute the expected value of X.
(e) (5 pts) Compute the variance of X.
(f) (5 pts) Compute the expected value and variance of Y = 2X+1.

Question 2 – 35 pts

A certain type of storage battery lasts on the average 3 years with a standard deviation of 0.5 year. Assuming the battery life X is normally distributed,

(a) (7 pts) find the probability that a given battery will last less than 2.3 years?
(b) (7 pts) find the probability that a given battery will last between 2 and 3 years?
(c) (7 pts) given the battery lasts more than a year, find the probability that a given battery will last more than 2 years?
(d) (7 pts) Find the value of T such that P(X>T) = 0.25.
(e) (7 pts) Find the value of T such that P(X>T) = 0.80.

Question 3 – (35 pts)

The length of time for one individual to be served at a cafeteria is a random variable having an exponentially distribution with a mean of 4 minutes.

(a) (5 pts) What is the probability that a person is served more than 5 minutes in a given day?
(b) (5 pts) What is the probability that a person is served between 3 and 5 minutes in a given day?
(c) (5 pts) Given than the person has not been served for 3 minutes, what is the probability that a person is served more than 5 minutes?
(d) (7 pts) What is the probability that a person is served in less than 3 minutes on at most 2 of the next 6 days?
(e) (5 pts) On the average, how many customers will be served in 10 minutes?
(f) (8 pts) What is the probability that exactly 2 customers are served in 10 minutes?